

Operating System:

An operating system is a software that communicates with the hardware and allows other programs to run.

- High level to low level language conversion
- used as intermediate

Types of OS:

1. Desktop OS (general purpose)
2. Server OS (remote accessible)



Windows



Windows 8



Linux



fedora



FreeBSD_®



ubuntu



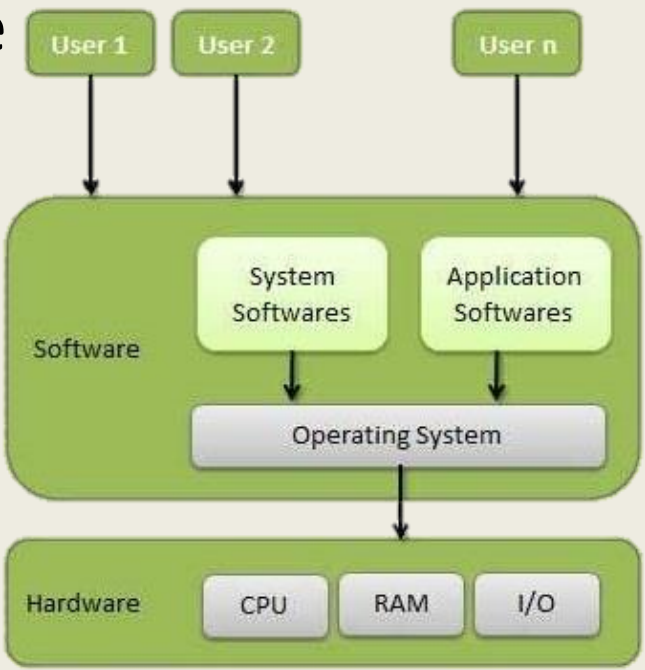
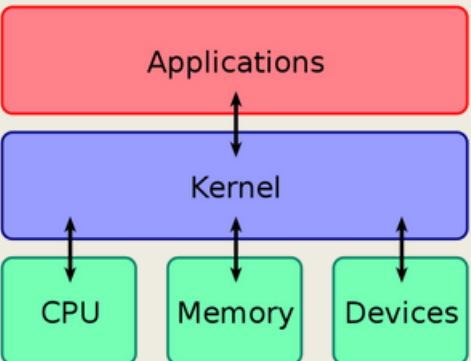
Mac OS



Android



- An operating system is a program that acts as an interface between the user and the computer hardware and controls the execution of all kinds of programs



- A kernel connects the application software to the hardware of a computer.

Development of OS:

- Single user with single tasking at same time.
- Single user with multiple tasking at same time.
- Multiple user with multiple tasking at same time.

History:

- C –language (Dennis Ritchie)-1972
- Unix OS (Ken Thompson and Dennis Ritchie -Bell Labs) -1973
 - open source
- IBM => AIX, HP => HPUX, Solaris => sun OS
 - paid services
- GNU Linux Project (Richard Stallman)
 - According to the GNU Linux project, the Linux operating system's kernel is Linux but all other elements of the system are GNU
- Linus Torvalds
 - Linux kernel -free operating system kernel (Sept. 1991)

Result : Linux Kernel + GNU = Linux OS

Linux :

- The Linux open source operating system, or Linux OS, is a freely distributable, cross-platform operating system based on Unix that can be installed on PCs, laptops, notebooks, mobile and tablet devices, video game consoles, servers, supercomputers and more...

Basis For Comparison

Linux

Windows Performance

Definition

-open-source operating system

-developed by Microsoft Company

Kernel

-MonolithicKernel based OS i.e. consume more running space

-Micro lithic Kernel based OS i.e. consume less running space with efficiency lower than Linux

Benefits

- open-source (source code is available)
- Free of cost
- developer can download and modify the kernel of it according to requirement can distribute

- Closed source
- paid operating system
- Developed and modified by farm

Updating Process

-not require

-up to date in time to time

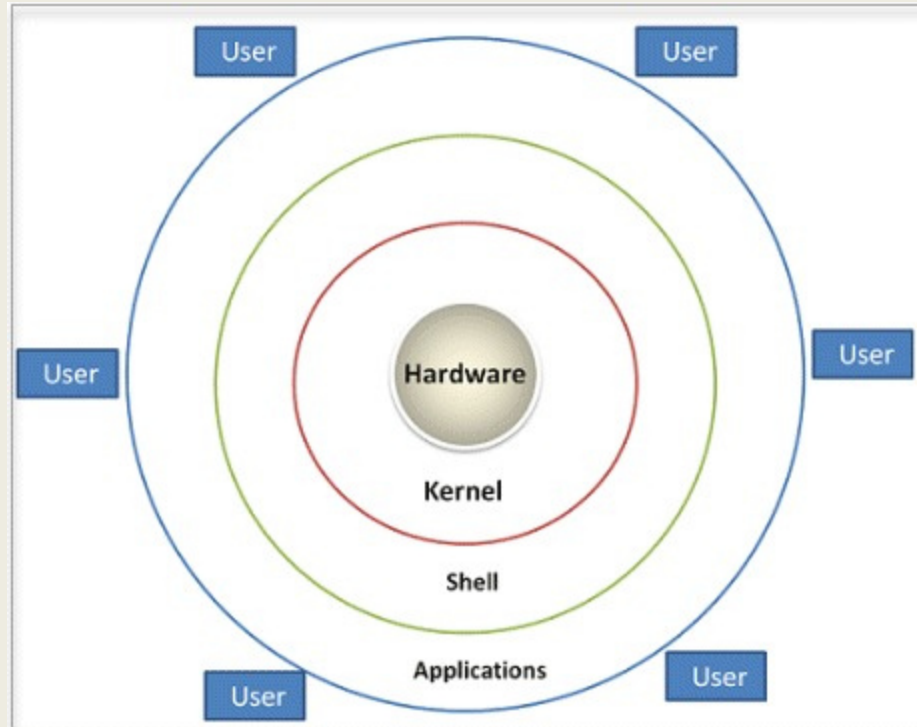
Software Made By

-Freely Available
Ex.Firefox,Thunderbird, VLC Media Player, Virtual Box etc.
-There are is lessAmount ofviruses in Linux compare towindows

-need licensed
Ex. MicrosoftOffice etc
-Many types of virus inwindows OS, Because of a large amount of market share

Security

Linux Kernel Architecture



Shell: The shell is command line interpreter, i.e. translator

It gathers input from user and executes programs kernel.

Bourne shell (sh), C shell (csh), Power shell , Kornshell (ksh), Bourne Again SHell(bash)

Kernel: The core component (heart) of an operating system.

- acts as a bridge between applications and the hardware.

- When an operating system is loaded into memory, the kernel loads first and remains in memory until the operating system is shut down again.

- responsible for low-level tasks such as disk mgmt., task mgmt. and memory mgmt.

- Monolithic Kernel, Micro Kernel, Hybrid Kernel, ExoKerne, Nano Kerne

- Installation process:
 - 1) oracleVirtualBoxor VM ware
 - for guest os
 - 2) OS -> RHEL 7, 7.1, 7.2, Cent OS

S.NO.	CLI	GUI
1.	CLI is difficult to use.	Whereas it is easy to use.
2.	Consumes low memory.	Consumes more memory.
3.	High precision.	Low precision is obtained.
4.	Faster than GUI.	GUI is slower than CLI.
5.	Needs only keyboard.	Need both mouse and keyboard.
6.	Appearance can not be modified or changed.	Appearance can be modified or changed.
7.	Input is entered only at command prompt.	Input can be entered anywhere on the screen.
8.	No graphics	Graphics are used.
9.	Spelling mistakes and typing errors are not avoided.	Spelling mistakes and typing errors are avoided.

LAB

Basic command :

- Shut down: init0
- Reboot : init6
- File : touch, cat, rm
- Directory : mkdir, rmdir
- Text : echo,
- Working dir. : pwd
- Change to home directory : cd
- running process : top
- Memory utilization : free ,du
- Force to remove directory : rm-rfv

Linux Users:

Root User	Local User	System User
• Super User • Root User • System Administrator	• normal user • regular user • Common users	• Systemuser • Server user
home dir. :/root	: /home/<users>	-
• By default created while installing	• created by command (#useradd)	• created by an os during installation which is used
• #	• \$	-
• No changes in root user name	• name can be suggested and must be unique.	-

```
CentOS Linux 7 (Core)  
Kernel 3.10.0-514.el7.x86_64 on an x86_64
```

```
localhost login: root  
Password:  
Last login: Tue Dec 11 08:02:31 on tty2  
[root@localhost ~]#
```

File Edit View Search Terminal Help

```
[sanjay@localhost ~]$
```

username

hostname

current
working
directory

[	username	@	hostname without domain name	last directory in current working directory path	]	User type. \$ for regular user and # for root user
---	----------	---	------------------------------------	--	---	--

hostname <NEW_HOSTNAME>

Command in Linux OS :

Syntax: command option(s) filename(s)

- Enter commands in lowercase.
- <Options> : modify the way in which command works, (-),
Multiple options in one command
- <Filename>: name of a file that you want to use
- spaces are must between commands, options, and filenames
- Multiple commands can written on same command line &
separated
by a semicolon (;)

Ex. \$ls, \$ls-l, \$ls-l /home/swapnil

Types of commands:

File Commands

- | | |
|--------------------|---|
| 1. ls | Directory listing |
| 2. ls-al | Formatted listing with hidden files |
| 3. ls-lt | Sorting the Formatted listing by time modification |
| 4. cd dir | Change directory to dir |
| 5. cd | Change to home directory |
| 6. mkdir dir | Creating a directory dir |
| 7. cat > file | Input into the file |
| 8. head file | Output the first 10 lines of the file |
| 9. tail file | Output the last 10 lines of the file |
| 10. rm file | Deleting the file |
| 11. rm-r dir | Deleting the directory Force to remove the file |
| 12. rm-f file | Force to remove the directory dir |
| 13. rm-rf dir | Copy the contents of file1 to file2 |
| 14. cp file1 file2 | |
| 15. cp-r dir1 dir2 | Copy dir1 to dir2;create dir2 if not present |
| 16. mv file1 file2 | Rename or move file1 to file2,if file2 is an existing directory |

Editors

1. Vi/Vim editor (required file to edit)
2. Cat editor (required file to edit) 3. Gedit editor
4. Nano editor (required file to edit) 5. GNU Emacs editor
6. Kate/Kwrite editor
7. Lime Text editor

Information Command

- | | | |
|----|-------------|--|
| 1. | lscpu | cpu details |
| 2. | lsusb lspci | usb details |
| 3. | dmidecode | peripheral component inter communication |
| 4. | free | all information |
| 5. | mandb | volume related information |
| 6. | | |

Helping Command

1. **info**
2. **man**
3. **help**
4. **whatis**

- grep:searches for PATTERNS in each FILE
grep<optn.> <wordstofind>[file]
- sort : Accenting/Desending(-r) order description
-

Redirector of cat:

-single redirector(>)

enter data as overwrite

-Double redirector(>>)

enter data as re-editing

-">&"

-">2" re-directs output of one file to another.

re-directs output of error in file

- Pipe (|)

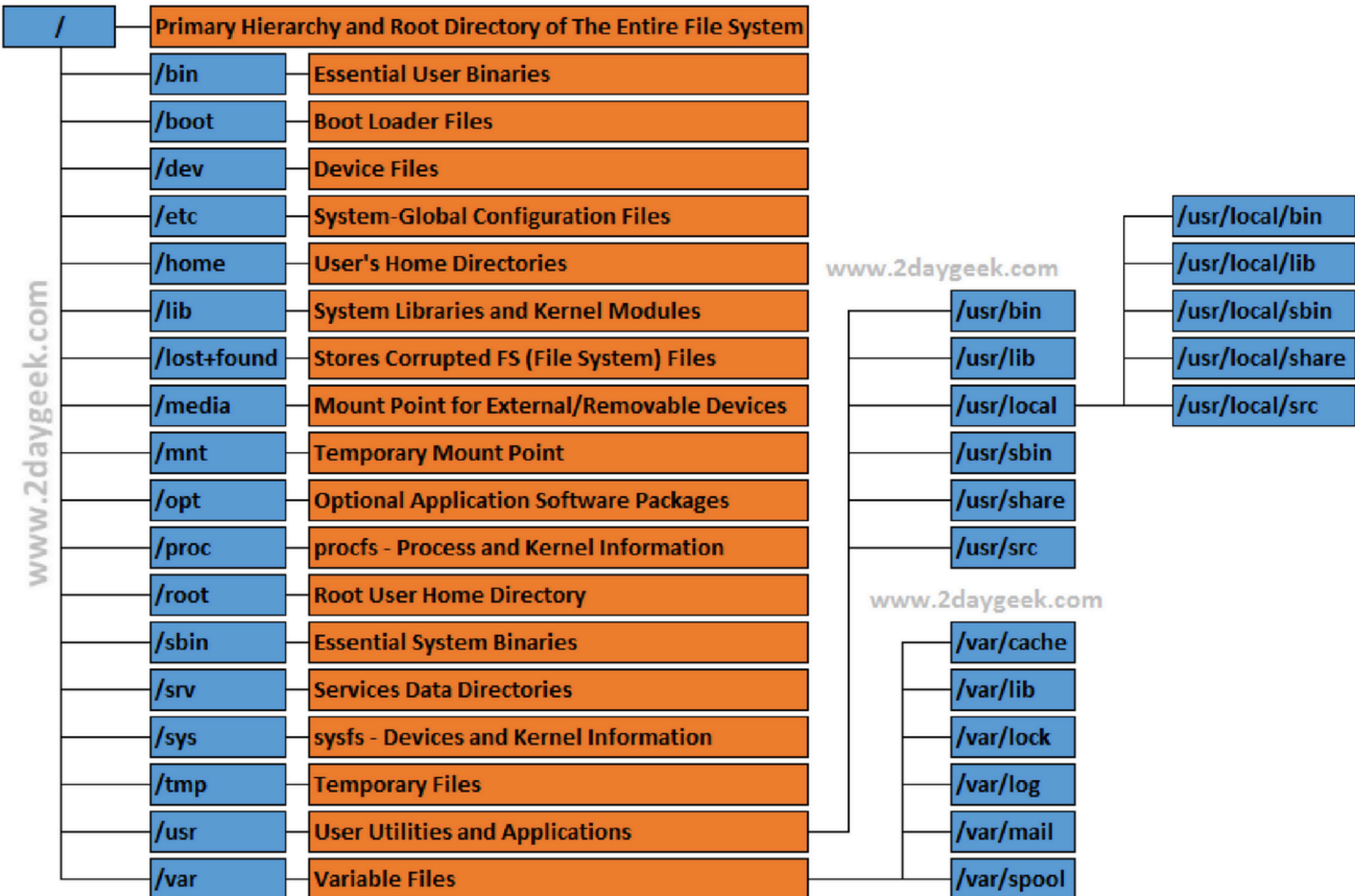
o/p of previous command as an i/p of next command

<comm.1> | <comm.2>

- Semicolon(;)

Multiple option performing

File system Hierarchy Standard :



User and Group Management :

- Types of users:
 - root user (uid-0)
 - system user (uid-1-999)
 - local user (uid-1000 –60,000+)
- Local user creation
 - useradd/ adduser<name of users>
- Password for local user
 - passwd<name of user>

- Basic files of user :



- ☐ uid& gid

- ☐ /home

- Skeleton files in home

- .bashrc, .bashprofile, .bashlogout

- /etc/skel/



- ☐ Var/spool/mail /<user>



- ☐ /etc/passwd



- ☐ /etc/shadow



- ☐ /etc/group



- ☐ /etc/gshadow

/etc/passwd



The diagram illustrates the structure of the `/etc/passwd` file. It features a dark red header bar with the text `/etc/passwd` in a green, monospace font. Below the header, a horizontal row of seven light brown boxes represents the fields of a password entry, separated by small black dots (colons). The fields are labeled as follows:

Username or login name	Encrypted password	User ID	Group ID	User description	User's home directory	User's login shell

1) Username 2)Linked Password /etc/shadow 3)User ID (UID) (-u) 4)Group ID (GID) (-g) 5)User ID Info (-c) 6)Home directory (-d) 7)Command/shell (-s) Syntax: `adduser-<optn.> <requ. of optn.> <user name>`

/etc/shadow

- 1) Username
- 2) Encrypted Password
- 3) Last password change (-d)
- 4) Minimum password age(-m)
- 5) Maximum password age (-M)
- 6) Warning period(-W)
- 7) Inactivity period(-I)
- 8) Expiration date. (-E)
- 9) Unused.

Syntax: chage-<optn.> <user name>

user:\$1\$.pj1GQnU\$ZWEt3P2/CpbqeiWGa/Ymd1:18556:0:99999:7:::

\$id\$

ID of the
algorithm
used

\$salt\$

Random data
that makes
password
unique

\$hash\$

Hash of
the user
password

Encrypted
password

Days since
epoch of last
password change

Minimum days until
password change allowed

Maximum days the
password is valid

Number of days before
warning for expiration

Number of days
since account
was locked

Number of days
without login
after expiration
before account
is locked

User login name

adduser,addgroup-add a user or group to the system

usermod:modify a user account

userdel, groupdel: remove a user or group to the system

chage, passwd: used to view and change the user password expiry information.

/etc/group

- 1) group_name
- 2) Password
- 3) Group ID (GID)
- 4) Group List

Syntax: gpasswd-<optn.> <group name>

/etc/gshadow

- 1)Group Name
- 2)EncryptedPasswordOf Group
- 3)Group Administrator
- 4)Group Members

Long listing (ll, ls-l):

- 1)The file type (7 types)
- 2)The file permissions (rwx)
- 3)Number of hard links to the file.
- 4)File owner.
- 5)File group.
- 6)File size.
- 7)Date and Time.
- 8)File name.

File types:

File type :	Description
Ordinary or regular files (-)	Contain data of various content types such as text, script, image, videos, etc.
Directory files (d)	Contain the name and address of other files.
Block (b) and character special files (c)	Represent device files such as hard drives, monitors, etc.
Link files (l)	Point or mirror other files
Socket files (s)	Provide inter-process communication
Named pipe files (p)	Allow processes to send data to other processes or receive data from other processes.

- 1) identifying the type of content that is saved in the file
- 2) supports seven different types of files
- 3) # file <path of file>

link count

- For file = 1
- ForDirectory = 2 (note: parent directory only = +1)

File link types:

- a. Hard link (-) ---syntax (#ln<exist file> <hl file>)
---replica, link count (+1)
---inodeis same
- a. Soft link (l) ---syntax (#ln-s <exist file> <slfile>)
---show path to original file, link count (+0)
---inodeis different

Inode Number

- In addition to its file name, each file in a file system has an identification number, called an inode number (that is unique in its file system)
- The inode number refers to the physical file, the data stored in a particular location.
- A file also has a device number, and the combination of its inode number and device number is unique throughout all the file systems in the hierarchical file system.

ls-i

File Permission Modes

- Owner permissions–
- Group permissions–
- Other permissions–

Types of permissions:

- read (r)
- write (w)
- execute (x)

Syntax: #Chmod<ugo> <+ -=> <rwX> <attributes>

Octal numbers :

Octal number	Binary number	r w x	Permissions:
0	000	- - -	none
1	001	- - x	execute only
2	010	-w -	write only
3	011	-w x	write and execute
4	100	r - -	read only
5	101	r - x	read and execute
6	110	rw -	read and write
7	111	rw x	read, write, and execute (full permissions)

Default permission for: 1) root

a) file :-644

b) directory :-755

2) user

a) file :-664

b) directory :-775

Umasknumber :

- The user file-creationmodemask (umask)
 - Determine file permission for newly created files
 - control thedefault file permission for new files.
 - It is a three-digit octal number
 - can be set or expressed using:
 - Symbolic values (r,w,x,u,g,o,a,=,-,+)
 - Octal values (001, 110)

Super user : 022

Localuser : 002

Conf. file: /etc/profile

Syntax: umask

Update : source/etc/profile

- Full permission for

- file:- 666

- Directory : 777

- Default permission

- » = FullPermission -umasknum.

Ownership

- to change the ownership and group ownership of a

#chown<user>:<group> <attributes>

chgrp <group> <attributes> or

chown:<group> <attributes>

Special permission

- Types of specialp.
 - setuidPermission (suid)
 - This special permission allows all users to access files and directories that are normally only available to root.
 - Applied on executable files only & replace 'x' by 's' (/sbin/dmidecode)
 - Syntax : #Chmodu+s<path of comm.>
 - setgid Permission (sgid)
 - directory belonging
 - Group property inherent
 - files that were created in this directory belong to the group
 - user who has write and execute permissions in the directory can create a file.
 - syntax : chmodg+s<dir>
 - StickyBit
 - directory belonging
 - file can be deleted only by the owner of the file, the owner of the directory, or byroot.
 - prevents a user files from deleting other users in group
 - syntax: chmodo+t <dir>

Access Control List

- Special permission for user / group
- access to files and/or directories.
- ACLs tell operating systems which users can access data, and what privileges the users are allowed.
- Syntax: `#setfacl-m u:<user name>:rwx`
 <dir/file>
 - `# getfacl`
 - `Setfacl -b/x <file/dir to remove acl/particular user or grp>`

Super user do (sudo)

- Local u.....comm.Ofroot.....operate
- /etc/suders.....(r—r--- ---)
- Visudo
- Wheel grp
.....user.....conf(comment)

Filters and search utility

- Grep
 - Pipe
 - Syntax: `#grep "<finding words>" <path/file>`
 - `-i`ignore casesensitive
 - `-E` ...multiple word finding Pipe
 - `-c`counting
 - `-n` ...positionoffline
- Locate
 - searchall system
 - `# locate <finding word>`
 - `# updatedb` (to update kernel)
- Find
 - Follows options to find
 - Syntax: `#find </> <-name> <findingwords>`
 - `-name` (find by name) `-type` (find by type of file) `-user` (user related/created files) `-exec` (execution on founded command)

Archive and compression

- Tape archive(tar)
 - for create and extract Archive
 - to create compressed or uncompressed Archive files
 - 2-5% compression
 - Syntax: tar[options][archive-file] [file or directory to be archived]
 - -c create, -f file, -v verbose -t listing -x extract
- Disk usage (du)-shows memory utilization (-sh)
- Compression
 - gzip(-z) .gzip.....65-70% compression
 - Ex: #tar cfz[archive and comp.-file] [file or directory to be archive and comp.]
 - xfz[file to extract] -C [Destination to uncompress. archive]
 - tar file (.tar) compression by : gzip, uncompress: gunzip
 - Bzip2(-j) .bz2
 - Options- cfj(compress archive), x fj(uncompress. archive)
 - For compression of tar file: bzip2 <tar file> ,uncompress: bunzip2
 - Xzip(-J) .xz.....70-80% compression
 - Options- cfJ(compress archive), x fJ(uncompress. archive)
 - For compression of tar file: xz<tar file> ,uncompress: unxz

Job Scheduling:

-to arrange for something to happen or be done at a particular time

Tools for scheduling-

- at
 - Non periodic task
 - Syntax: #at "<time>"<schedule task>
 - atq (to see task), atrm <n> (to remove task)
- Cron tab(/etc/crontab)
 - periodic task
 - Single time with multi job performing
 - Required running server
 - Option : -l (schedule), -e (editing previous task),
- Anacron
 - Single time job execution
 - work on offline server

- Cron:

- Allowed special character (*, -, /, ?, #)
- Asterik(*)–Match all values in the field or any possible value.
- Hyphen(–)–To define range.
- Slash (/)–1st field /10 meaning every ten minute or increment of range.
- Comma (,)–To separate items..

Package Management

- Bundles offfile
- Install, managing, maintenance, remove,configuration
- Repository consist packages . (like playstore)
- Redhat family (centOS, RHEL, Fedora)
 - A) low level tool..... Rpm
 - options: -q(installed or not), -qa(all pkgs.), -ivh<full pkgname>
 - B) High level tool.... yum
- Debian family(ubuntu)
 - A) low level tool..... dpkg
 - B) High level tool..... apt
- Downloading comm.
 - wget
 - curl -O

- rpm (redhatpackage management)
 - Process:
 - Copybinary link -download –install –use
 - Options:
 - -q(query), -qa(quarry all), -qi(information), -evh(remove pkg),
-qip(completpkginfo) –ql(list content of pkg.)
- Yum (yellowdogupdate manager)
 - Syntax:#yum<options> <pkg.name>
 - Install, remove, autoremove(including dependencies), provides
(pkgdetails o/p with help of command), search, list installed,
reinstall (to recover deleted command fromrepo.),
 - Yumdownloader<pkgname> (to download pkg)

Repository

- Ex. Play store
- Yum.repo.d
- Steps to create repository:
 - Directory ☐ packages
 - Reopdata# createrepo<path of stored pkg.>
 - Configuration ☐ /etc/yum.repos.d/<conf. file of repository>.repo
 - [abc] (repo ID/Repository)
 - (name ID)
 - name=<abc_1>
 - baseurl=
 1. http://
 2. ftp://
 3. file://(complete path of store pkg. dir.)
 - Enabled=1,0
 - Gpgcheck=0 (permission)
 - # yum clean all (to notify kernel)
 - # yum repolist (unable repository list)

Partition Mgmt./Disk Mgmt.

- Divide a single hard drive into many logical drives using partition editors like fdisk, parted tools.
- Directories and files of different categories can be stored in different partitions.
- Partitions are represented by device files, located in /dev.
-

Data transmission hard drive type: -PATA & SATA

- bus interface used for connecting secondary storage devices like hard disks, optical drives, CD drive etc.
 - PATA (Parallel Advanced Technology Attachment)
 - hda
 - PATA cables are used with internal drives (46 cm)
 - Req. large space
 - 40-conductor ribbon cables
 - replacement or addition of components to system without stopping, shutting down or rebooting the system is not possible
 - SATA (serial Advanced Technology Attachment)
 - rotary hard drives with spinning platters and a moving head that writes data to consecutive sectors on each platter
 - Sda, sdb
 - 7-conductor ribbon cables
 - Faster running process than PATA
 - supports hot swapping

How to differentiate between SATA and PATA



SATA drive
(has card-edge connector)

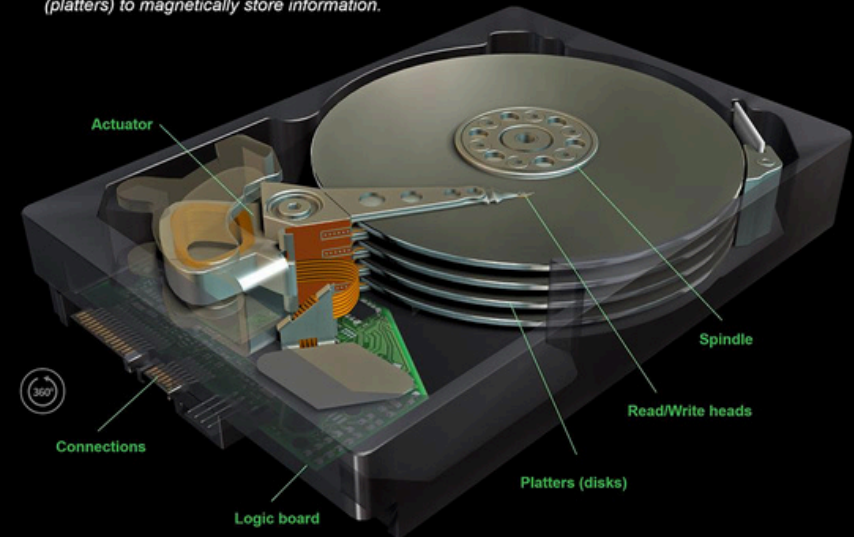


PATA drive
(has pin connector)

How Hard Disk Drives Work

Presented by SEAGATE.COM
Designed & research by ANIMAGRAFFS

Hard Disk Drives (HDDs) use spinning disks (platters) to magnetically store information.



Presented by SEAGATE.COM • Design & research by ANIMAGRAFFS